Edward-Constantin Scutariu

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EDUCATION

Babes-Bolyai University — **Cluj-Napoca, Romania**
B.Sc in Computer Science — *October 2023 - Present*

COURSES

Databases, Data Structures, Operating Systems, OOP, Artificial Intelligence, Web Programming

EXPERIENCE

CloudFlight — **August 2025 - October 2025**
Training programme at Cloudflight in partnership with UBB — *Cluj-Napoca, Romania*

- Developed a full-stack web application for managing ingredients and personal fridge items, implementing **REST APIs**, CRUD operations, and reactive UI navigation.
- Built features for adding, editing, deleting, and filtering ingredients, including a user-specific **"My Fridge"** module for nutritional information management.
- Implemented data persistence and database migrations using **Spring Boot, Flyway, and MySQL**, ensuring schema stability and data validation.
- Integrated **Swagger** for interactive API documentation and testing, with validation and secure data handling across the front-end and back-end.
- Followed modern development practices, including **Agile/Scrum**, **Git** version control, automated testing concepts, and **CI/CD** pipeline fundamentals.

PROJECTS

RUBLUT

React | Three.js | Spring Boot | BLE

- Full-stack web application for interactive 3D Rubik's Cube solving and learning via Bluetooth Low Energy (BLE) connectivity.
- **Authentication & Persistence –** Secure user registration, authentication, and state automatic persistence in a **MySQL** database.
- **Real-time Hardware Sync –** Integrates Web Bluetooth API to connect smart cubes (GoCube) and display physical turns instantly on a 3D model using React Three Fiber.
- **Intelligent Solvers –** Provides step-by-step guidance (first move hint) and full auto-solving algorithm (Kociemba) integrated via **Spring Boot** backend.
- **Interactive Configuration & Tutorials –** Allows manual color mapping for standard cubes and features an educational tutorial module (LBL, CFOP, Roux, WCA notation).

Pathfinding Visualizer

Java | JavaSwing | JFrame

- A **Java Swing** application based on **JFrame**, allowing the visualization of pathfinding algorithms in an interactive matrix.
- **Intuitive interface –** Users can set the grid size, start/end points, and add obstacles.
- **Pathfinding algorithms –** Supports **Lee**, **DFS**, **BFS**, and **A*** for finding the shortest path.
- **Maze generation –** Includes algorithms like **Binary Tree** and **Backtracking** for automatic maze.
- **Control and customization –** Adjust execution speed, quick reset, and a simple navigation menu.

TrackForUBB

CSharp | React | TypeScript

- During a team project, developed TrackForUBB, a full-stack university management platform using **React + TypeScript** and **ASP.NET Core with Entity Framework** and **PostgreSQL**, covering grades, timetables, exams, and contracts.
- Designed a secure RESTful API with layered architecture and **Swagger/OpenAPI** documentation.
- Implemented role-based authentication and authorization using **JWT** and **Microsoft Entra** (Azure AD).
- Developed advanced features including timetable and subject management, PDF contract generation from templates, CSV import/export, and a modular React architecture with reusable components.
- Contributed to developing key frontend modules: exam scheduling pages, grade management (view/place grades), locations & classrooms CRUD, FAQ component, improved error handling, and loading states for backend data fetching.

EXTRACURRICULAR ACTIVITIES

Hackathons

- I participated in two hackathons, and even though I didn't contribute much to the coding part, I learned a lot about effective teamwork and gained valuable experience.

Cloudflight Coding Contest

- I had the pleasure of participating in 4 Competitive Programming CCC contests as part of a team, placing in the top 100 in 2 contests, and in the top 200 in 2 others.

KEY SKILLS

Programming Languages C++, Java, Python, SQL
Technologies Qt, JavaFX